

YVONNE YU

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EDUCATION

New York University

M.S. in Data Science, Courant School of Math, Computing & Data Science (GPA: 3.72/4.0) Sept. 2025 – May 2027

Sun Yat-sen University

Bachelor of Economics, Lingnan College (GPA: 3.80/4.0) Sept. 2021 – Jun. 2025

RESEARCH & PROJECTS

Seeing Sentiment: News Images, Retail Participation & Chinese A-Share Returns Nov. 2024 – Jun. 2026

- **Research Design & Data:** Tested whether news images add return-predictive information beyond text; built a MongoDB corpus of **245K articles/572K images** (2014–2026), aligned with 5 A-share indices over 2,982 trading days.
- **Sentiment Measurement:** Fine-tuned ViT-B/16 in PyTorch/Hugging Face (**88.1%** held-out PCNN accuracy); paired image scores with Chinese RoBERTa text scores to construct daily visual and textual pessimism indices.
- **Econometric Analysis:** Estimated Newey–West regressions in R with volatility subsamples and text orthogonalization; **84%** of image-signal variance remained unexplained by text, with incremental predictability on high-volatility ChiNext days.
- **Robustness & Interpretation:** Tested liquidity, news-volume, and lagged-return controls plus a volume placebo; compared effects against 3 retail-density proxies, finding the strongest image-sentiment association in retail-intensive ChiNext.
- **Out-of-Sample Validation:** Replicated expanding-window forecasts in statsmodels with past-only scaling; across 1,467 forecasts, ChiNext achieved $R^2_{OOS} = \mathbf{0.293\%}$ versus the historical mean (Clark–West $p = 0.015$).

Retrieval-Coordinated Reasoning for Small LLMs Feb. 2026 – May 2026

- **Benchmark Design:** Evaluated **7 models and 10 strategies** across ReClor, LogiQA 2.0, CommonsenseQA, and ConTRoL using identical prompts and official validation splits to test when small-model teaming improves reasoning.
- **Failure Diagnosis:** Applied paired McNemar tests and bootstrap intervals; voting, debate, and judging hurt accuracy on 3 short-context benchmarks ($p < 0.05$), with failures linked to correlated errors and miscalibrated confidence.
- **Pipeline Design:** Built a pipeline pooling premise evidence from 3 heterogeneous retrievers for one answerer; compared cross-model majority checks with single-judge filtering, using direct-answer fallback if no evidence passed.
- **Validation & Audit:** F1 (including fallback) raised Qwen3-8B ConTRoL accuracy from **65.5% to 71.7% (+6.2 pp)** vs. single-model retrieval; LettuceDetect flagged **< 0.02%** of retrieved-evidence characters as unsupported across 805 cases.

INDUSTRY EXPERIENCE

ByteDance

Data Scientist Intern, E-Commerce Search May 2026 – Aug. 2026

- **Product & Query Research:** Compared placement/query/lifecycle impression–click–purchase funnels with manual demand and q2q/i2q candidates; revealed merchant store/brand-query coverage gaps and appliances/3C cold-start priorities. Rebuilt supporting Module/Query datasets in SQL/Spark.
- **Repeat-Search Causal Analysis:** Estimated search/content/mall effects on T+2–T+7 active repeat search via user-level DML-SparseLinear and 30-day controls; a reversal A/B test found repeat-search frequency and probability below control at high exposure, guiding recall and exit rules.

Mars

Data Scientist Intern May 2025 – Aug. 2025

- **Feature & Model Pipeline:** Built resource-day telemetry/cost features; screened idle assets with Isolation Forest and ranked candidates with XGBoost using time splits, reaching about **85% Precision@100** in manual review.
- **Deployment & Impact:** Deployed calibrated Azure ML scoring with a 14-day appeal/feedback loop; verified decommissioning/right-sizing yielded annualized savings and cut investigations from **52 to 31 minutes (-40%)**.

Korn Ferry

Data Science Consultant Intern Aug. 2024 – Dec. 2024

- **Compensation Data Product:** Built Python ETL for pay/financial data, harmonizing currency, year, job family, industry, and region into a traceable query layer; cut annual pay review from **two weeks to two days (-80%)**.
- **Talent Segmentation:** Standardized 28 pay/performance/tenure features across employee records; PCA reduced them to 6 dimensions and K-Means formed **4 clusters**, informing retention, succession, and job-pricing recommendations.